

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: A Centralized Platform for Educational and Career Opportunities

GROUP NUMBER and MEMBERS: Group No: 5 / Members: Kaan DÖNMEZ, Hilal Seray ORTAN, Baran Deniz AYDIN, Salih Batuhan ŞENER, Doruk YAŞAR, Selin TAY, Ömer ŞEN

Questions to identify measurements:

- Did each sprint finish according to the planned schedule?
- How much effort did the team spend on each feature and sprint?
- How many tasks were completed in each sprint?
- How many defects were found during testing?
- How effective were the test cases for the main features?
- Did the system performance improve after optimization?
- How many requirement or feature changes occurred during the project?
- Were important quality areas such as security and response time completed before the final sprint ended?

Identified measurements:

- Planned sprint / actual completion sprint for each feature
- Planned task completion date / actual task completion date
- Number of completed tasks per sprint
- Weekly development hours and weekly testing hours
- Number of defects found and number of critical defects
- Total test cases / passed test cases / failed test cases
- Test pass rate = $\text{passed test cases} / \text{total test cases} \times 100$
- Number of requirement or feature changes
- Average system response time before and after optimization
- Number of open and closed issues in the project tracking tool

Measurement storage and collection:

- **What:** Planned and actual completion information for Sprint 1, Sprint 2, Sprint 3, and Sprint 4
When: At the end of each sprint
Format: Sprint name, planned date, actual date, completed/not completed
How: Recorded in the project schedule or shared spreadsheet by the project manager
- **What:** Development and testing hours
When: At the end of each week
Format: Integer or real number data
How: Each team member records hours in a shared spreadsheet
- **What:** Defects, critical defects, and change requests
When: Whenever a problem or change is found
Format: Integer counts and short descriptions
How: Recorded in GitHub Issues, Trello, or a defect/change log
- **What:** Test case results and response time
When: After each testing session and after optimization work
Format: Total, passed, failed test cases and response time values
How: Recorded by the testing member in the testing spreadsheet

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Measurement Type	Description	Example Measurements
Schedule	Compares planned sprint/date with actual completion sprint/date.	User Registration and Authentication: planned in Sprint 1; actual completion sprint/date will be recorded.
Sprint Progress	Measures how many planned tasks are completed in each sprint.	Sprint 2: Search & Filter, Personalized Feed, Tracking, Bookmark, and Profile tasks completed / planned tasks.
Effort	Measures time spent on development, testing, meetings, and fixes.	Weekly hours spent on AI-Based Recommendation Engine in Sprint 3.
Quality / Defects	Measures product quality by tracking defects and critical defects.	Number of defects found in Notification System, Deadline Reminder System, or Security Implementation.
Test Pass Rate	Shows the percentage of successful test cases.	Search & Filter Functionality: passed test cases / total test cases x 100.
Change Data	Tracks requirement or feature changes during the project.	Number of changes made to Personalized Feed System or Application and Tryout Tracking after feedback.
Performance	Measures system speed and response time before and after optimization.	Average response time before and after Performance & Response Time Optimization in Sprint 4.